

Literature Status for arXiv:1006.1546: Intersection-Body Fixed Points

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Status

This is a `literature_already_answered` packet. It records that the fixed-point question for the intersection body operator quoted in Kim–Yaskin–Zvavitch, *The geometry of p -convex intersection bodies*, arXiv:1006.1546, is answered by Milman–Shabelman–Yehudayoff, *Fixed and Periodic Points of the Intersection Body Operator*, arXiv:2408.08171.

Source Question

On PDF page 2 of arXiv:1006.1546, the authors recall a question about the intersection body operator I , citing Lutwak and Gardner. Since a Euclidean ball is mapped by I to a Euclidean ball, they ask whether there are other fixed points of I . They note that Fish–Nazarov–Ryabogin–Zvavitch had proved only a local uniqueness result in a sufficiently small Banach–Mazur neighborhood of the ball, and that the general question was still open.

The same paragraph then asks whether IK must be closer to the Euclidean ball than K . That second, distance-improvement question is answered negatively in arXiv:1006.1546 itself; it is not the literature-status claim of this packet.

Supporting Answer

The abstract of arXiv:2408.08171 states that, when $n \geq 3$,

$$I^2K = cK \quad \text{if and only if} \quad K \text{ is a centered ellipsoid,}$$

and hence

$$IK = cK \quad \text{if and only if} \quad K \text{ is a centered Euclidean ball.}$$

On PDF page 2, Theorem 1.1 gives the I^2 classification and Corollary 1.2 gives the fixed-point classification for I . The authors explicitly say that Corollary 1.2 provides a complete answer to Gardner’s fixed-point question and to the Fish–Nazarov–Ryabogin–Zvavitch question about fixed points of $I : \mathcal{S}_n \rightarrow \mathcal{S}_n$ when $n \geq 3$.

Scope

This answers the fixed-point branch quoted in the source paper for star bodies in dimensions $n \geq 3$. The two-dimensional case is exceptional: the later paper notes that $I^2K = 4K$ for every origin-symmetric star body in $\mathbb{R}^2$, and $IK = 2K$ for bodies invariant under a quarter-turn. The source

paper’s separate question about whether IK improves Banach–Mazur distance to the Euclidean ball is a same-paper negative result and is not counted here as a separate literature answer.

Search Evidence

The lane-3 queue selected arXiv:1006.1546. Cheap run indexes were searched for the arXiv id, 2408.08171, intersection body fixed points, p-convex intersection bodies, Lutwak, and Gardner; no duplicate packet for this source relation was found. Local corpus search and targeted web search for fixed points intersection body operator identified arXiv:2408.08171 as the decisive later source.

References

- [1] J. Kim, V. Yaskin, and A. Zvavitch, *The geometry of p -convex intersection bodies*, arXiv:1006.1546.
- [2] E. Milman, S. Shabelman, and A. Yehudayoff, *Fixed and Periodic Points of the Intersection Body Operator*, arXiv:2408.08171.